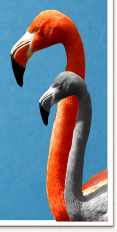
CHAPTER 2 Modeling Distributions of Data

2.1 Describing Location in a Distribution

The Practice of Statistics, 5th Edition
Starnes, Tabor, Yates, Moore



Bedford Freeman Worth Publishers

Compare Data Sets

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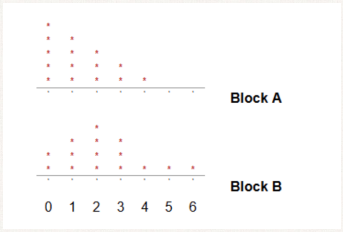
When you compare two or more data sets, focus on four features:

•**Center**. Graphically, the center of a distribution is the point where about half of the observations are on either side.
•**Spread**. The spread of a distribution refers to the variability of the data. If the observations cover a wide range, the spread is larger. If the observations are clustered around a single value, the spread is smaller.
•**Shape**. The shape of a distribution is described by symmetry ,skewness, number of peaks, etc.
•**Unusual features**. Unusual features refer to gaps (areas of the distribution where there are no observations) and **outliers**.

Dotplots

When dotplots are used to compare data sets, they are positioned one above the other, using the same scale of measurement, as shown below.

Question: The dotplots show pet ownership in homes on two city blocks. Interpret the dotplots. Each dot represents a household.



- **Shape:** Symmetry, Number of peaks (mode) , Skewness, Uniform
- **Center:** Median
- **Spread:** Range, Variance, SD, IQR
- **Outliers/Gaps**

Stem-and-Leaf Plots

Question: The weight of two groups of eight animals, Group M and Group N, are recorded and the data is shown in the stem plots below (with each stem and leaf representing weight in kg). Use the stem plots to compare the weight of the animals in the two groups.

Group M:

1	4
2	3 4 8
3	2 6 8
4	
5	0

Group N:

1	0
2	3 6
3	5
4	1
5	4 7
6	2

- **Shape:** Symmetry, Number of peaks (mode) , Skewness, Uniform
- **Center:** Median
- **Spread:** Range, Variance, SD, IQR
- **Outliers/Gaps**

Describing Location in a Distribution

Learning Objectives

After this section, you should be able to:

- ✓ FIND and INTERPRET the percentile of an individual value within a distribution of data.
- ✓ ESTIMATE percentiles and individual values using a cumulative relative frequency graph.
- ✓ FIND and INTERPRET the standardized score (z-score) of an individual value within a distribution of data.
- ✓ DESCRIBE the effect of adding, subtracting, multiplying by, or dividing by a constant on the shape, center, and spread of a distribution of data.

Measures of Position for Quantitative Data

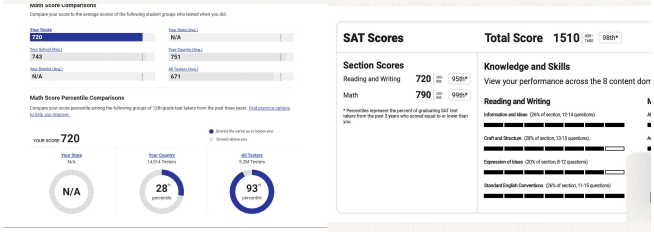
Statisticians often talk about the position of a value, relative to other values in a set of quantitative data.

The most common measures of position are percentiles 百分位数, quartiles 四分位数, and standard scores (aka, z-scores) 标准计分.

The interquartile range (IQR) is a measure of variability, based on dividing a data set into **quartiles**.

Quartiles divide a rank-ordered data set into four equal parts. The values that divide each part are called the first, second, and third quartiles; and they are denoted by Q1(25%), Q2(50%), and Q3(75%), respectively.

Measuring Position: Percentiles



Measuring Position: Percentiles

One way to describe the location of a value in a distribution is to tell what percent of observations are less than it.

The pth percentile of a distribution is the value with p percent of the observations less than it. 将一组数据从小大大排列，第p百分位数

Here are the scores of all 25 students in Mr. Pryor's statistics class on their first test:

79 81 80 77 73 83 74 93 78 80 75 67 73
77 83 **86** 90 79 85 83 89 84 82 77 72

The bold score is Jenny's 86. How did she perform on this test relative to her classmates?

6	7
7	2334
7	5777899
8	00123334
8	569
9	03

Key: 7|2 is a student who scored 72 on the test

Cumulative Relative Frequency Graphs

There are some interesting graphs that can be made with percentiles. One of the most common graphs starts with a frequency table for a quantitative variable.

For instance, the frequency table in the margin summarizes the ages of the first 44 U.S. presidents when they took office.

Age	Frequency
40–44	2
45–49	7
50–54	13
55–59	12
60–64	7
65–69	3

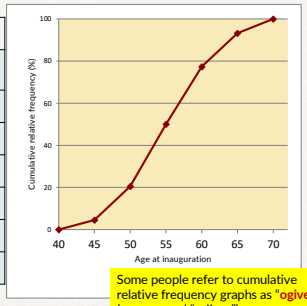
Age of First 44 Presidents When They Were Inaugurated

Age	Frequency	Relative frequency	Cumulative frequency	Cumulative relative frequency
40-44	2	2/44 = 4.5%	2	2/44 = 4.5%
45-49	7	7/44 = 15.9%	9	9/44 = 20.5%
50-54	13	13/44 = 29.5%	22	22/44 = 50.0%
55-59	12	12/44 = 34%	34	34/44 = 77.3%
60-64	7	7/44 = 15.9%	41	41/44 = 93.2%
65-69	3	3/44 = 6.8%	44	44/44 = 100%

Example

Age of First 44 Presidents When They Were Inaugurated

Age	Frequency	Relative frequency	Cumulative frequency	Cumulative relative frequency
40-44	2	2/44 = 4.5%	2	2/44 = 4.5%
45-49	7	7/44 = 15.9%	9	9/44 = 20.5%
50-54	13	13/44 = 29.5%	22	22/44 = 50.0%
55-59	12	12/44 = 34%	34	34/44 = 77.3%
60-64	7	7/44 = 15.9%	41	41/44 = 93.2%
65-69	3	3/44 = 6.8%	44	44/44 = 100%



Some people refer to cumulative relative frequency graphs as "ogives" (pronounced "o-jives").

(a) Was Barack Obama, who was first inaugurated at age 47, unusually young?

(b) Estimate and interpret the 65th percentile of the distribution.

Measuring Position: Z-Scores

A z-score tells us how many standard deviations from the mean an observation falls, and in what direction.

Z-score 告诉你一个数据点的**relative position**。如果Z-分数为正，表示该数据点高于平均值；如果为负数，表示低于平均值。Z-分数的绝对值越大，表示与平均值的偏差越大。

Z-score 还可以用于检测**outlier** 。通常情况下，如果Z-分数超过了一定的阈值，则可以认为该数据点是一个异常值，因为它与平均值的偏差较大。

If x is an observation from a distribution that has known mean and standard deviation, the standardized score of x is:

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

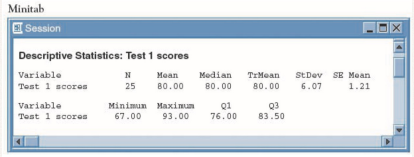
A standardized score is often called a z-score.

Using z-scores for comparisons

Let's return to the data from Mr. Pryor's first statistics test. Where does Jenny's score of 86 fall relative to the mean of this distribution/ standardized score (z-score)?

6	7
7	2334
7	5777899
8	00123334
8	569
9	03

Key: 7|2 is a student who scored 72 on the test



$$z = \frac{x - \text{mean}}{\text{standard deviation}} = \frac{86 - 80}{6.07} = 0.99$$

One day after, Jenny earned an 82 on Mr. Goldstone's chemistry test. At first, she was disappointed … Then Mr. Goldstone told the class that the distribution of scores was fairly symmetric with a mean of 76 and a standard deviation of 4.

Problem: On which test did Jenny perform better relative to the class? Justify your answer.

Transforming Data

Transforming converts the original observations from the original units of measurements to another scale. Transformations can affect the shape, center, and spread of a distribution.

Effect of Adding (or Subtracting) a Constant

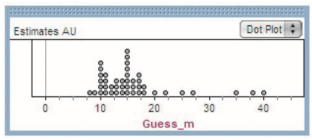
Adding the same number *a* to (subtracting a from) each observation:

Example

Soon after the metric system was introduced in Australia, a group of students was asked to guess the width of their classroom to the nearest meter. Here are their guesses in order from lowest to highest

8	9	10	10	10	10	10	10	11	11	11	11	12
12	13	13	13	14	14	14	15	15	15	15	15	15
15	15	16	16	16	17	17	17	17	18	18	20	22
25	27	35	38	40								

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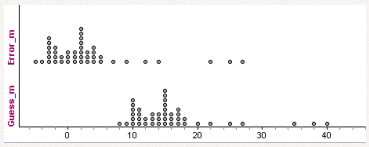
	n	$\bar{x}$	s_x	Min	Q_1	Med	Q_3	Max	IQR	Range
Guess	44	16.02	7.14	8	11	15	17	40	6	32

Example : Transforming Data

Examine the distribution of students' guessing errors by defining a new variable as follows:

error = guess − 13

That is, we'll subtract 13 from each observation in the data set. Try to predict what the shape, center, and spread of this new distribution will be.



	n	Mean	Sx	Min	Q1	M	Q3	Max	IQR	Range
Guess(m)	44	16.02	7.14	8	11	15	17	40	6	32
Error (m)	44	3.02	7.14	-5	-2	2	4	27	6	32

Transforming Data

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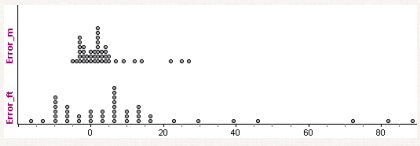
Effect of Adding (or Subtracting) a Constant

Adding the same number *a* to (subtracting a from) each observation:

- adds *a* to (subtracts *a* from) measures of center and location (mean, median, quartiles, percentiles), but
- Does not change the shape of the distribution or measures of spread (range, IQR, standard deviation).

Transforming Data

Because our group of Australian students is having some difficulty with the metric system, it may not be helpful to tell them that their guesses tended to be about 2 to 3 meters too high. Let's convert the error data to feet before we report back to them. There are roughly 3.28 feet in a meter.



	n	Mean	s_x	Min	Q_1	M	Q_3	Max	IQR	Range
Error (m)	44	3.02	7.14	-5	-2	2	4	27	6	32
Error (ft)	44	9.91	23.43	-16.4	-6.56	6.56	13.12	88.56	19.68	104.96

Transforming Data

Transforming converts the original observations from the original units of measurements to another scale. Transformations can affect the shape, center, and spread of a distribution.

Effect of Multiplying (or Dividing) by a Constant

Multiplying (or dividing) each observation by the same number *b*:

- multiplies (divides) measures of center and location (mean, median, quartiles, percentiles) by *b*
- multiplies (divides) measures of spread (range, IQR, standard deviation) by $|b|$, but
- does not change the shape of the distribution

Effects of Changing Units

	Center (mean, median)	Spread (range, IQR, standard deviation)
Adding a Constant c	Adding c	Unchanged
Multiply by a Constant c	Multiply by c	Multiply by $\|c\|$

Describing Location in a Distribution

Section Summary

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